Dheeraj Dhillon

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EDUCATION

University of Wisconsin-Madison **Aug 2025 - May 2027**
Master of Science (M.S.), Computer Science
Coursework: Reinforcement Learning, Machine Learning, Natural Language Processing

Indian Institute of Technology (IIT) Roorkee **Jul 2019 - May 2023**
Bachelor of Technology (B.Tech.), Electronics and Communication Engineering

WORK EXPERIENCE

Gartner Inc., Associate Data Scientist, Gurgaon **Mar 2024 - Aug 2025**

- Built and productionized a 3-level hierarchical topic-modeling pipeline over 30K+ Gartner research documents and 20K+ client inquiries using all-MiniLM-L6-v2 embeddings, FAISS similarity graphs, and Louvain community detection to surface temporal demand trends, support supply-demand forecasting, and facilitate fine-grained inquiry-to-document matching.
- Led implementation of the logistic regression successor to Gartner's production Random Forest Content Implied Value Rating (IVR) model, enabling confidence intervals, broader document coverage, and greater interpretability; reconciled LR/RF IVR outputs through decile-level comparisons to support business adoption.
- Scaled Content IVR to 21K+ documents across 17 contract types and 100K+ licensed users by adding a document-read flag to contract-specific retention models, fitting one MLE logistic regression per document–contract pair, and measuring read-vs.-not-read probability deltas; parallelized fitting with Spark RDDs across 24 workers and deployed monthly workflows.
- Developed Attribute IVR to identify drivers of document retention performance, regressing IVR on 200+ topic-profile features, embedding-based uniqueness, engagement recency, dwell time, and job-role composition; identified 50+ statistically significant drivers (adjusted $R^2 = 0.62$ on 4K IT End User documents) that informed author content strategy.

HiLabs Inc., Data Scientist, Bangalore **Jul 2023 - Mar 2024**

- Extended the Python/OOP business-rules layer of a production Medicare/Medicaid provider-roster ingestion pipeline, translating client requirements into SQL CRUD updates and richer conditional logic across state, NPI, TIN, and related fields to enable more granular, configurable processing.
- Implemented an R&D proof of concept to recover metadata discarded outside roster tables, training a custom spaCy NER model on synthetically generated KEY/VALUE examples and combining entity predictions with delimiter and cell-proximity heuristics to surface candidate values for missing standardized fields during downstream review.

Microsoft R&D, Data and Applied Scientist Intern, Hyderabad **May - Jul 2022**

- Conducted an NLP/IR feasibility study for message autosuggestions in Microsoft Teams Search, benchmarking 8 embedding-, statistical-, and graph-based keyphrase extractors—including KeyBERT, YAKE, TextRank, and TopicRank—across 6 datasets totaling 1,050 documents; KeyBERT achieved 30.6% F1@5 vs. 25.6% for YAKE on 500 Inspec abstracts.

PROJECTS

ReAct Agent for Multi-Hop Question Answering on HotpotQA **Jul - Aug 2026**

- Built a ReAct multi-hop QA agent in LangGraph with Qwen2.5-7B served via vLLM on the 7,405-question HotpotQA FullWiki dev set, combining BM25 and dense retrieval, BGE reranking, and sentence-level evidence memory to search and reason across Wikipedia in up to 7 actions.
- Used retrieved bridge evidence to formulate subsequent search and lookup actions for multi-hop questions, outperforming a single-pass RAG baseline that passed the 7 highest-ranked Wikipedia pages to Qwen2.5-7B; improved exact match from 40.5% to 46.7% and answer F1 from 51.8% to 60.5%.

Reinforcement Learning from Human Feedback (RLHF) for LLM Alignment **May - Jun 2026**

- Executed an end-to-end post-training RLHF pipeline on Qwen2.5-0.5B-Instruct using 36K HelpSteer3 preference pairs: LoRA SFT, a Bradley–Terry reward model (65.6% held-out preference accuracy), and KL-regularized PPO with TRL.
- Stabilized long-form generation up to 768 tokens through quantile reward clipping, EOS-aware scoring, and 4-gram repetition shaping; evaluated across 2,017 held-out prompts spanning General, STEM, Code, and Multilingual domains.

Trust Region Policy Optimization and Policy Gradient Methods, UW-Madison **Mar - Apr 2026**

- Implemented TRPO and PPO from scratch in PyTorch and benchmarked Gaussian MLP and categorical CNN policies across 3 MuJoCo locomotion tasks and 7 Atari control environments; PPO achieved higher Hopper and Walker2d returns with 10× fewer environment steps than the TRPO training setup.
- Connected the theoretical progression from Conservative Policy Iteration and Natural Policy Gradients to TRPO and PPO through surrogate performance bounds and policy-space constraints.
- Ablated Hopper NPG with 3× and 9× fixed step sizes, showing the trade-off between learning speed and policy-update stability and motivating TRPO's KL-constrained step selection.

SKILLS

Programming Languages: Python, C++, SQL, C
ML, RL & LLM Tools: PyTorch, Hugging Face Transformers, TRL, PEFT, Stable-Baselines3, Gymnasium, vLLM, LangGraph
Simulation & Data Tools: CARLA, MuJoCo, Arcade Learning Environment (ALE), FAISS, Databricks, PySpark